

ADAM PARKOSIDIS

Science Park 904, 1098XH
Anton Pannekoek Institute for Astronomy
University of Amsterdam
Amsterdam, North Holland, The Netherlands

EDUCATION

Master of Science <i>Physics and Astronomy</i> 'Cum Laude'	2021 – 2023
University of Amsterdam-Vrije Universiteit Amsterdam, The Netherlands	
• Thesis: <i>Mass Transfer in Hierarchical Triple Systems: The Case of Roche-Lobe filling Outer Star</i>	
Bachelor of Science <i>Physics</i>	2013 – 2020
Aristotle University of Thessaloniki, Greece	
• Thesis: <i>Dynamic Perturbations in Protoplanetary Cosmic Dust Discs in the Presence of a Giant Planet: Possible Detection with ALMA</i>	

PUBLICATIONS

Reframing the wide eccentric binary problem: Eccentricity as a probe of mass-transfer physics	2026, accepted ApJ
A. Parkosidis, S. Toonen, E. Laplace, V. Schaffenroth	
Rethinking mass transfer: a unified semianalytical framework for circular and eccentric binaries. II. Orbital evolution due to non-conservative mass transfer	2026, A&A
A. Parkosidis, S. Toonen, E. Laplace, F. Dosopoulou	
Rethinking mass transfer: a unified semianalytical framework for circular and eccentric binaries. I. Orbital evolution due to conservative mass transfer	2026, A&A
A. Parkosidis, S. Toonen, F. Dosopoulou, E. Laplace	
Episodic outbursts during brown dwarf formation	2025, A&A
A. Parkosidis, D. Stamatellos, B. Riaz	

SCIENTIFIC EXPERIENCE

PhD Candidate in Astrophysics	2023 – Present
Anton Pannekoek Institute, UvA, The Netherlands	
Technical Support Scientist	2023
ASTRON, Dwingeloo, The Netherlands	
• Supported LOFAR Unified Calculator for Imaging (LUCI) tool development by testing, debugging, and integrating new Python functionalities	
• Developed observation and processing templates utilizing LOFAR's Telescope Manager Specification System following proposal approval	
• Authored documentation and user guides on instrument capabilities and operational procedures	
• Performed quality control of data products generated by the Long-Term Archive (LTA) pipelines, producing reports and performance summaries	
Research Intern in Computational Astrophysics	2021
Jeremiah Horrocks Institute, UCLan, United Kingdom	
• Applied the PHANTOM smoothed particle hydrodynamics (SPH) code for large-scale numerical simulations of protoplanetary disks	
• Utilized the UCLan High-Performance Computing (HPC) cluster for parallel execution and resource-optimized SPH model runs	
• Processed, analyzed, and visualized simulation data using Python (NumPy, pandas, matplotlib, seaborn) and SPLASH	
Space Debris Data Analyst	2020 – 2023
SpaceDot, Thessaloniki, Greece	
• Collaborated with 80+ engineers and scientists to integrate debris mitigation data into mission design	

- Performed orbit propagation and collision analyses with ESA's DRAMA software to ensure compliance with international space debris mitigation standards
- Developed Python scripts for automating orbital simulations, data analysis, and visualization of on-orbit collision risk scenarios
- Contributed technical content to the System Design and Mission Requirements (SDMR) and Orbital Collision Risk Assessment (OCRA) documents for the Critical Design Review (CDR) phase

FUNDING & GRANTS

Leids Kerkhoven-Bosscha Fonds (LKBF)

Years granted: 2024, 2025, 2026

TALKS

<i>When Mass Transfer Makes Orbits Eccentric: Connecting Theory and Observation</i>	2026
European Astronomical Society	Lausanne, CHE
<i>Eccentricity as a Probe of Mass-transfer Physics</i>	2026
Colloquium: Institute of Astronomy	KU Leuven, BEL
<i>Mass Transfer in Eccentric Binaries: A Unified Framework & New Insights</i>	2025
Binary Stars in a New Era	Lijiang, CHN
<i>Mass Transfer in Eccentric Binaries: A Unified Framework & New Insights</i>	2025
Netherlands Astronomy Conference	Nijmegen, NL

TECHNICAL SKILLS

Operation Systems: Linux (Ubuntu), Microsoft Windows
Programming: Python, Bash, C, F77/F95
Machine Learning & Deep Learning: PySR, Keras, Pandas, PyTorch, Scikit-Learn, TensorFlow
Data Visualization: Matplotlib, Seaborn, Plotly, gnuplot
Document Creation: Microsoft Office Suite, LibreOffice Suite, Google Suite, LaTeX
Other Technologies: Smoothed Particle Hydrodynamics (SPH), Git

SCIENTIFIC COMPUTING

SPH codes: PHANTOM, SEREN
Stellar Evolution codes: MESA
N-body codes: SWIFT, HUAYNO
Python Packages: NumPy, SciPy, SymPy, DISKLAB, Astropy, PICASO
Space Debris Risk Assessment & Mitigation Analysis: DRAMA
Other software: AMUSE, Mathematica

TEACHING EXPERIENCE & STUDENT SUPERVISION

University Lectures	2026
• 'Convection' (M.Sc. course: 'Structure & Evolution of Stars')	
Summer projects	2024
• Magdalena Andrea Vilaxa Campos "Modeling Mass Transfer Between Stars" (MSc level)	
Teaching Assistant	2022 – 2025
Anton Pannekoek Institute, UvA, The Netherlands	
• 'Atmospheres & Radiative Transfer' (M.Sc. course, 4 times)	
• 'Structure & Evolution of Stars' (M.Sc. course, 2 times)	
• 'Disks & Accretion' (M.Sc. course)	
AUTh, Greece	2018
• 'Observational Astronomy' (BSc course)	
Private Tutor	2018 – 2019
Thessaloniki, Greece	
• Tutored and prepared secondary education students for their final exams in the school subjects of Physics and Math	

TRAININGS & SCHOOLS

Binary compact objects on eccentric orbits and their gravitational-wave signatures Corfu, Greece	2026
ICESUN Summer School 2025: Binary Stars and Compact Objects Yunnan Observatories, China	2025
MESA Summer School KU Leuven, Belgium	2025
Nordic Winter School on Multimessenger Astrophysics Skeikampen, Norway	2024
Mastering your PhD University of Amsterdam, The Netherlands	2024

COLLABORATION MEMBERSHIPS

The Binarity at LOw Metallicity (BLOeM)	2026 – Present
Laser Interferometer Space Antenna (LISA)	2025 – Present

LANGUAGES

Greek: Native
English: Fluent
Dutch: Elementary
German: Elementary

VOLUNTEERISM

Guide and telescope operator Observatory of AUTH	2018 – 2019
Operated the optical telescope at the Aristotle University of Thessaloniki Observatory during visits by primary, junior high, and high school students.	